

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
plt.rcParams['figure.figsize'] = (10,6)

# -----
# 1. Import the datasets
# -----

covid_data = pd.read_csv("D:/Sx Varun/Sx Code/MCA/2nd_Tri/python/dataSet/covid_data.csv")
covid = pd.read_csv("D:/Sx Varun/Sx Code/MCA/2nd_Tri/python/dataSet/covid.csv")

# Check top rows
print("covid_data.csv:")
print(covid_data.head())
print("\ncovid.csv:")
print(covid.head())
```

```
covid_data.csv:
   Unnamed: 0    date      region  confirmed  active  \
0           0  27-03-2020  Andaman and Nicobar Islands      1      1
1           1  28-03-2020  Andaman and Nicobar Islands      6      6
2           2  29-03-2020  Andaman and Nicobar Islands      9      9
3           3  30-03-2020  Andaman and Nicobar Islands      9      9
4           4  31-03-2020  Andaman and Nicobar Islands     10     10
```

```
   cured  deaths
0       0       0
1       0       0
2       0       0
3       0       0
4       0       0
```

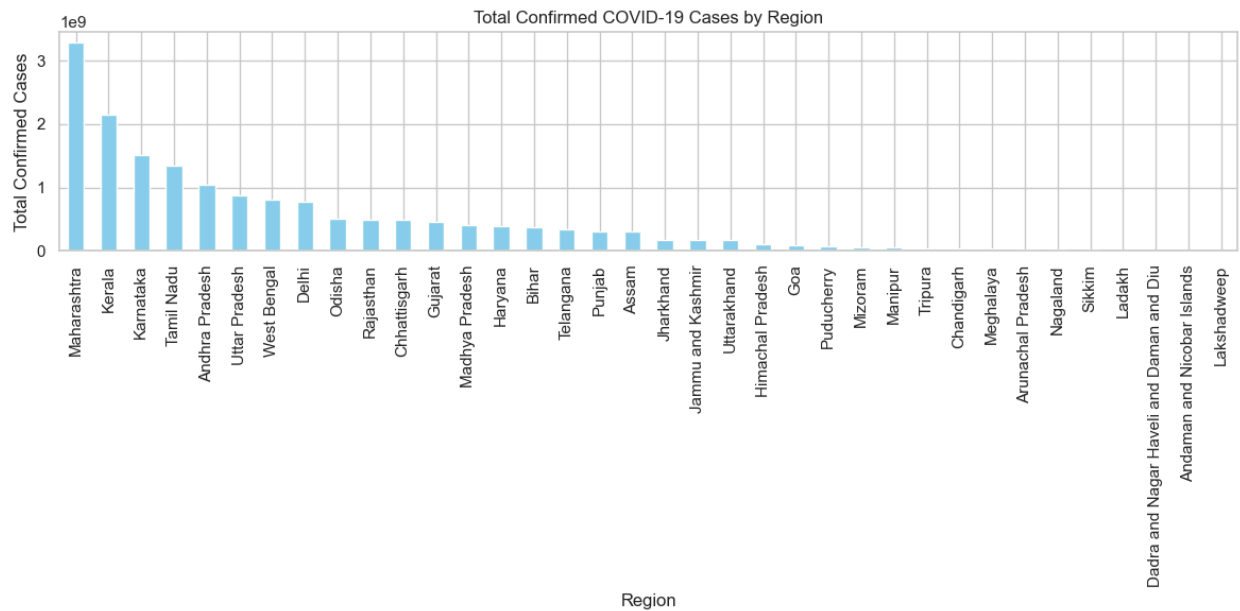
```
covid.csv:
   state  confirmed  active  passive  deaths  dose1  \
0  Andaman and Nicobar      10742      1     10612     129    313284
1    Andhra Pradesh     2339067      3    2324331    14733   40643161
2  Arunachal Pradesh     66890      0     66594     296    860442
3        Assam      746100      0     738065     8035   22549957
4        Bihar      851379     15     839062    12302   62944633

   dose2  dose3  precaution_dose  total_doses  population
0    320383   236936           53427      991263      426251
1  43549055  11703273       6579565    110556756    52883163
2    747177    72403         58618     1911760     1528296
3   20561790   2082670      1259853     50284713    34586234
4   59144387  11983504      3868082    157197041   119461013
```

```
In [2]: # -----
# 2. Bar Plot: Total Confirmed Cases by Region (from covid_data)
# -----

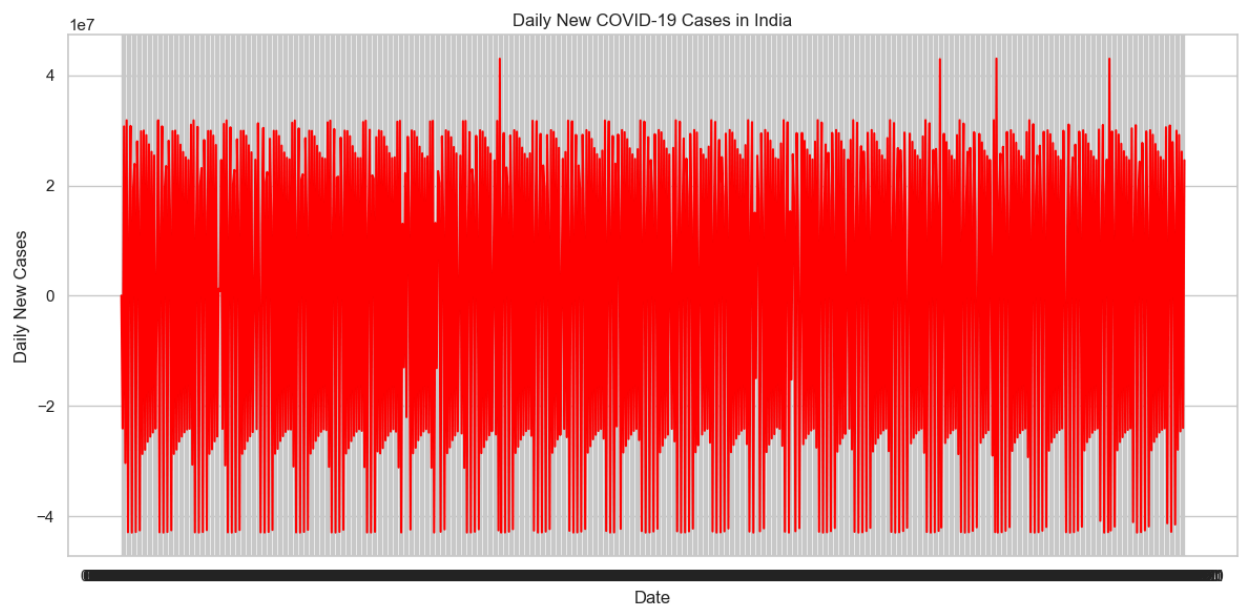
total_confirmed_by_region = covid_data.groupby('region')['confirmed'].sum().sort_values(ascending=False)

plt.figure(figsize=(12,6))
total_confirmed_by_region.plot(kind='bar', color='skyblue')
plt.title('Total Confirmed COVID-19 Cases by Region')
plt.xlabel('Region')
plt.ylabel('Total Confirmed Cases')
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```

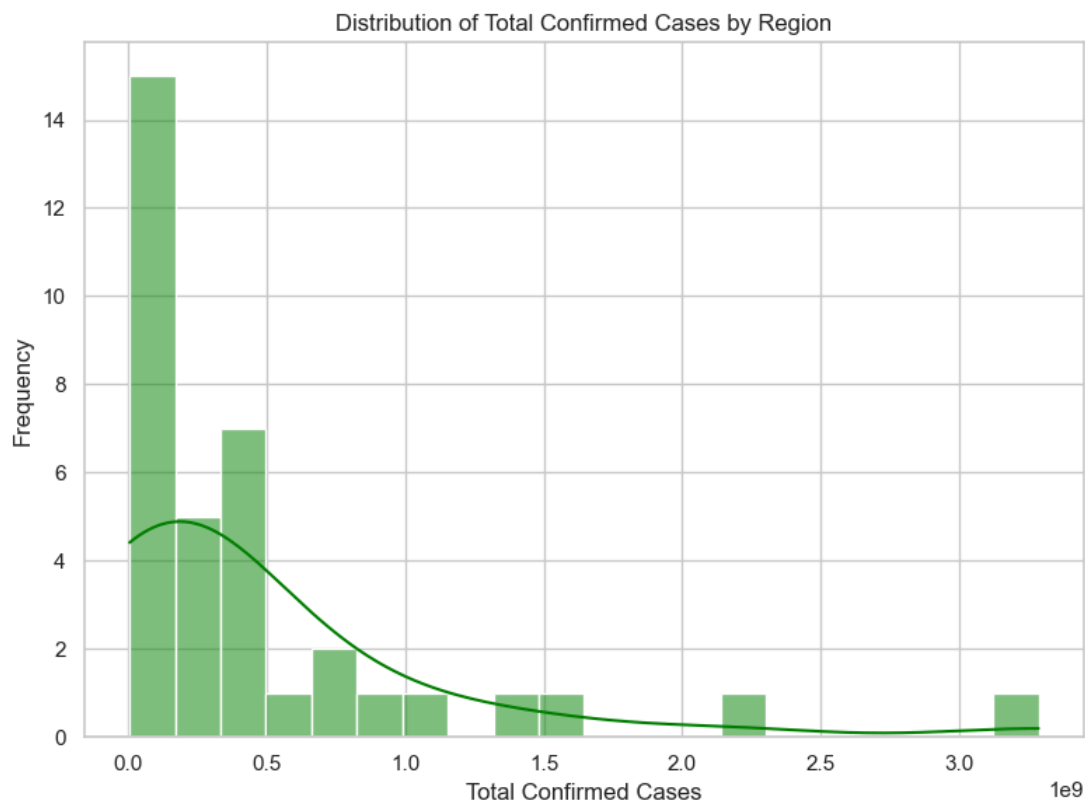


```
In [3]: # -----
# 3. Line Chart: Daily New Cases Over Time (from covid_data)
# -----
daily_new_cases = covid_data.groupby('date')['confirmed'].sum().diff().fillna(0)

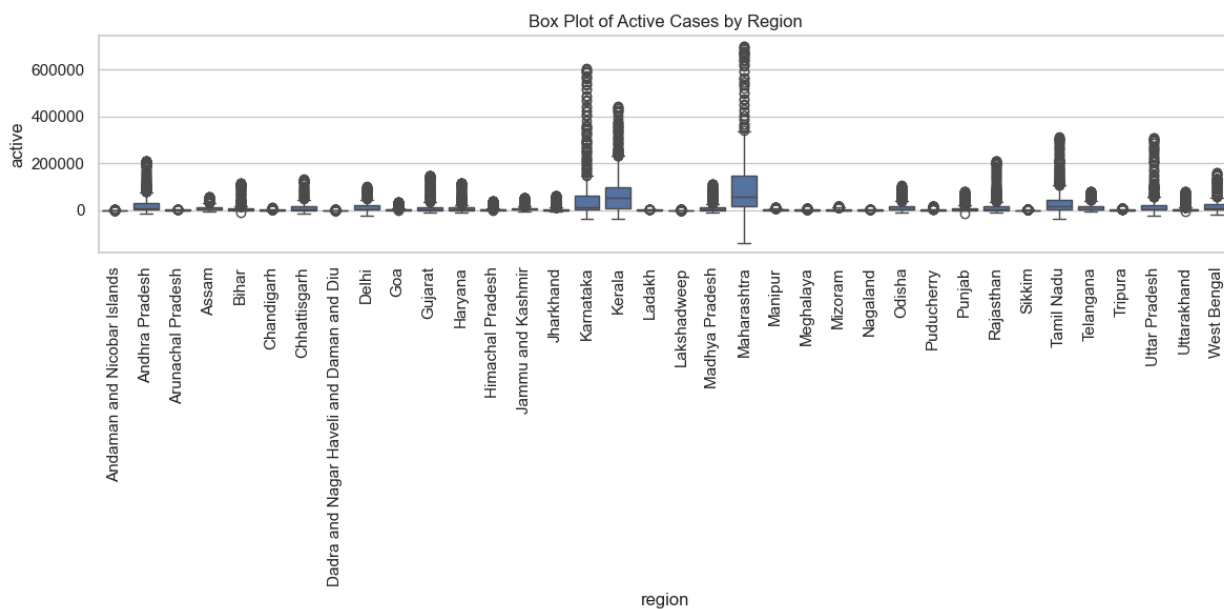
plt.figure(figsize=(12,6))
plt.plot(daily_new_cases.index, daily_new_cases.values, color='red')
plt.title('Daily New COVID-19 Cases in India')
plt.xlabel('Date')
plt.ylabel('Daily New Cases')
plt.tight_layout()
plt.show()
```



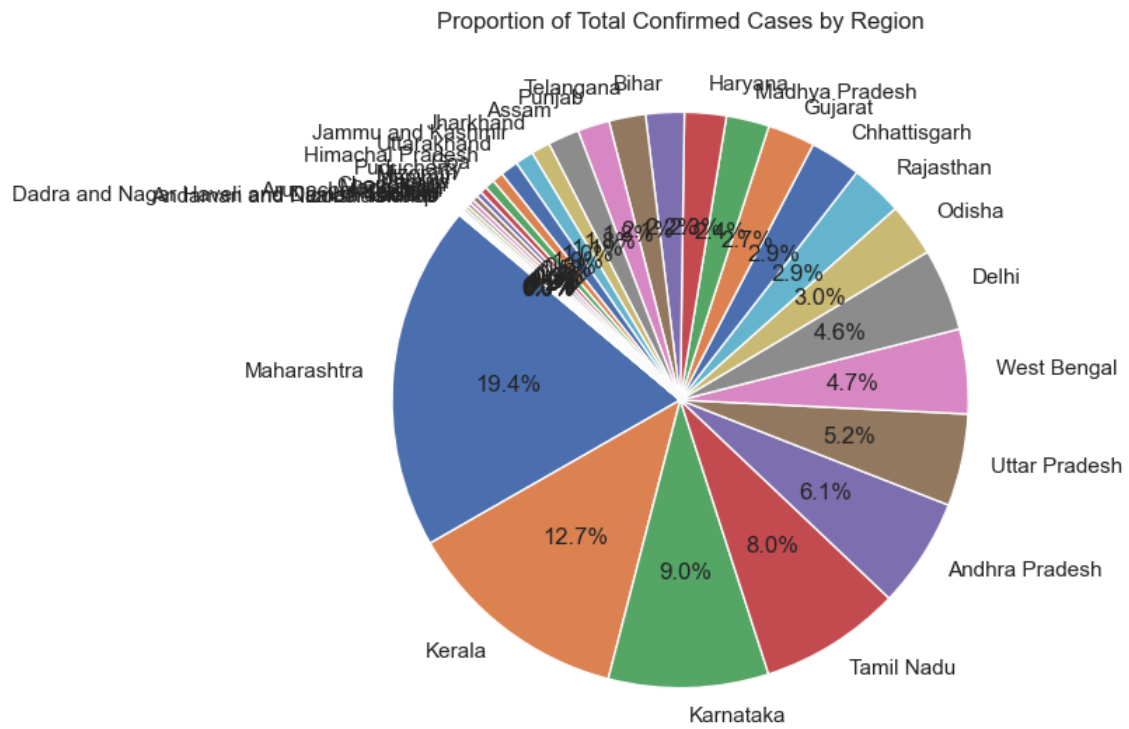
```
In [4]: # -----
# 4. Histogram: Distribution of Total Confirmed Cases by Region
# -----
plt.figure(figsize=(8,6))
sns.histplot(total_confirmed_by_region, bins=20, color='green', kde=True)
plt.title('Distribution of Total Confirmed Cases by Region')
plt.xlabel('Total Confirmed Cases')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



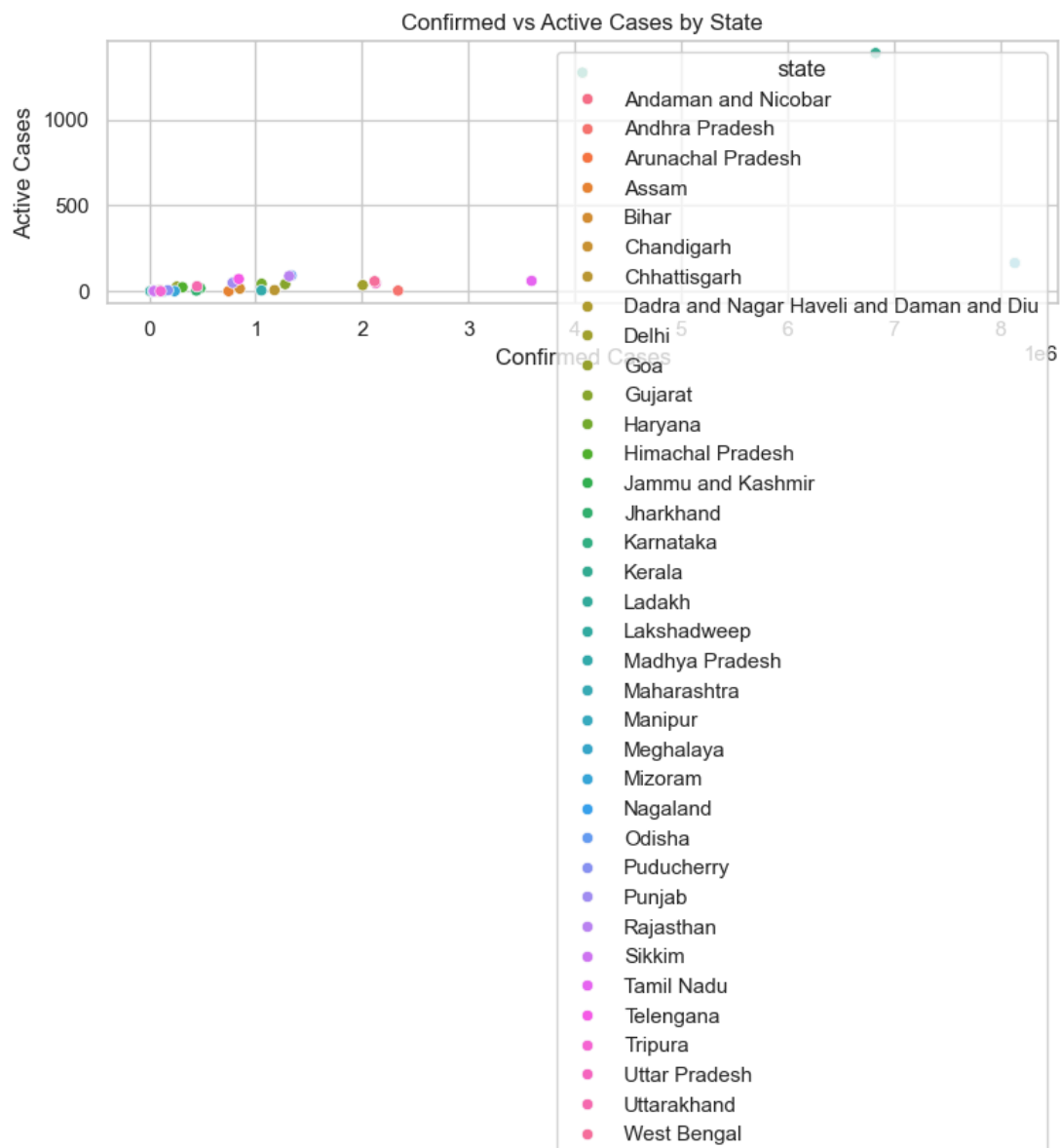
```
In [5]: # -----
# 5. Box Plot: Active Cases by Region (from covid_data)
# -----
plt.figure(figsize=(12,6))
sns.boxplot(x='region', y='active', data=covid_data)
plt.title('Box Plot of Active Cases by Region')
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



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In [6]: # -----
# 6. Pie Chart: Proportion of Total Confirmed Cases by Region
# -----
plt.figure(figsize=(8,8))
plt.pie(total_confirmed_by_region,
        labels=total_confirmed_by_region.index,
        autopct='%1.1f%%', startangle=140)
plt.title('Proportion of Total Confirmed Cases by Region')
plt.tight_layout()
plt.show()
```



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In [7]: # -----
# 7. Scatter Plot: Confirmed vs Active Cases (from covid.csv)
# -----
plt.figure(figsize=(8,6))
sns.scatterplot(x='confirmed', y='active', hue='state', data=covid)
plt.title('Confirmed vs Active Cases by State')
plt.xlabel('Confirmed Cases')
plt.ylabel('Active Cases')
plt.tight_layout()
plt.show()
```



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In [8]: # -----
# 8. Pair Plot (from covid.csv)
# -----
# Pick relevant numeric columns for pairplot
numeric_cols = ['confirmed', 'active', 'passive', 'deaths', 'dose1', 'dose2', 'dose3']
sns.pairplot(covid[numeric_cols])
plt.suptitle('Pair Plot of COVID-19 Metrics (States)', y=1.02)
plt.show()
```

Pair Plot of COVID-19 Metrics (States)

